```
!pip install gensim
```

```
Collecting gensim
  Downloading gensim-4.4.0-cp312-cp312-manylinux_2_24_x86_64.manylinux_2_28_x86_64.whl.metadata (8.4 kB)
Requirement already satisfied: numpy>=1.18.5 in /usr/local/lib/python3.12/dist-packages (from gensim) (2.0.2)
Requirement already satisfied: scipy>=1.7.0 in /usr/local/lib/python3.12/dist-packages (from gensim) (1.16.3)
Requirement already satisfied: smart_open>=1.8.1 in /usr/local/lib/python3.12/dist-packages (from gensim) (7.5.0)
Requirement already satisfied: wrapt in /usr/local/lib/python3.12/dist-packages (from smart_open>=1.8.1->gensim) (2.1.1)
Downloading gensim-4.4.0-cp312-cp312-manylinux_2_24_x86_64.manylinux_2_28_x86_64.whl (27.9 MB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 27.9/27.9 MB 18.6 MB/s eta 0:00:00
Installing collected packages: gensim
Successfully installed gensim-4.4.0
```

```
# Loading embeddings
import gensim.downloader as api

# Matrix handling
import numpy as np

# Visualization
import matplotlib.pyplot as plt

# t-SNE reduction
from sklearn.manifold import TSNE
```

```
# Load model
model = api.load("glove-wiki-gigaword-100")

# Print vocabulary size
print("Vocabulary Size:", len(model))

# Display one example vector
print("\nVector for word 'king':\n")
print(model['king'])
```

```
[==================================================] 100.0% 128.1/128.1MB downloaded
Vocabulary Size: 400000

Vector for word 'king':

[-0.32307  -0.87616   0.21977   0.25268   0.22976   0.7388   -0.37954
 -0.35307  -0.84369  -1.1113   -0.30266   0.33178  -0.25113   0.30448
 -0.077491 -0.89815   0.092496 -1.1407   -0.58324   0.66869  -0.23122
 -0.95855   0.28262  -0.078848  0.75315   0.26584   0.3422   -0.33949
  0.95608   0.065641  0.45747   0.39835   0.57965   0.39267  -0.21851
  0.58795  -0.55999   0.63368  -0.043983 -0.68731  -0.37841   0.38026
  0.61641  -0.88269  -0.12346  -0.37928  -0.38318   0.23868   0.6685
 -0.43321  -0.11065   0.081723  1.1569    0.78958  -0.21223  -2.3211
 -0.67806   0.44561   0.65707   0.1045    0.46217   0.19912   0.25802
  0.057194  0.53443  -0.43133  -0.34311   0.59789  -0.58417   0.068995
  0.23944  -0.85181   0.30379  -0.34177  -0.25746  -0.031101 -0.16285
  0.45169  -0.91627   0.64521   0.73281  -0.22752   0.30226   0.044801
```

```
 -0.83741   0.55006  -0.52506  -1.7357    0.4751   -0.70487   0.056939
 -0.7132    0.089623  0.41394  -1.3363   -0.61915  -0.33089  -0.52881
  0.16483  -0.98878 ]
```

```
words = [
    # Animals
    "dog", "cat", "lion", "tiger", "elephant", "wolf", "monkey", "horse",

    # Fruits
    "apple", "banana", "mango", "orange", "grape", "pineapple", "papaya", "peach",

    # Cities
    "london", "paris", "tokyo", "delhi", "mumbai", "newyork", "berlin", "rome",

    # Technology
    "computer", "laptop", "mobile", "keyboard", "internet", "software", "hardware", "robot",

    # Royal / People
    "king", "queen", "prince", "princess", "man", "woman", "boy", "girl"
]
```

```
word_vectors = []

for word in words:
    word_vectors.append(model[word])

word_vectors = np.array(word_vectors)

print("Shape of word vectors:", word_vectors.shape)
```

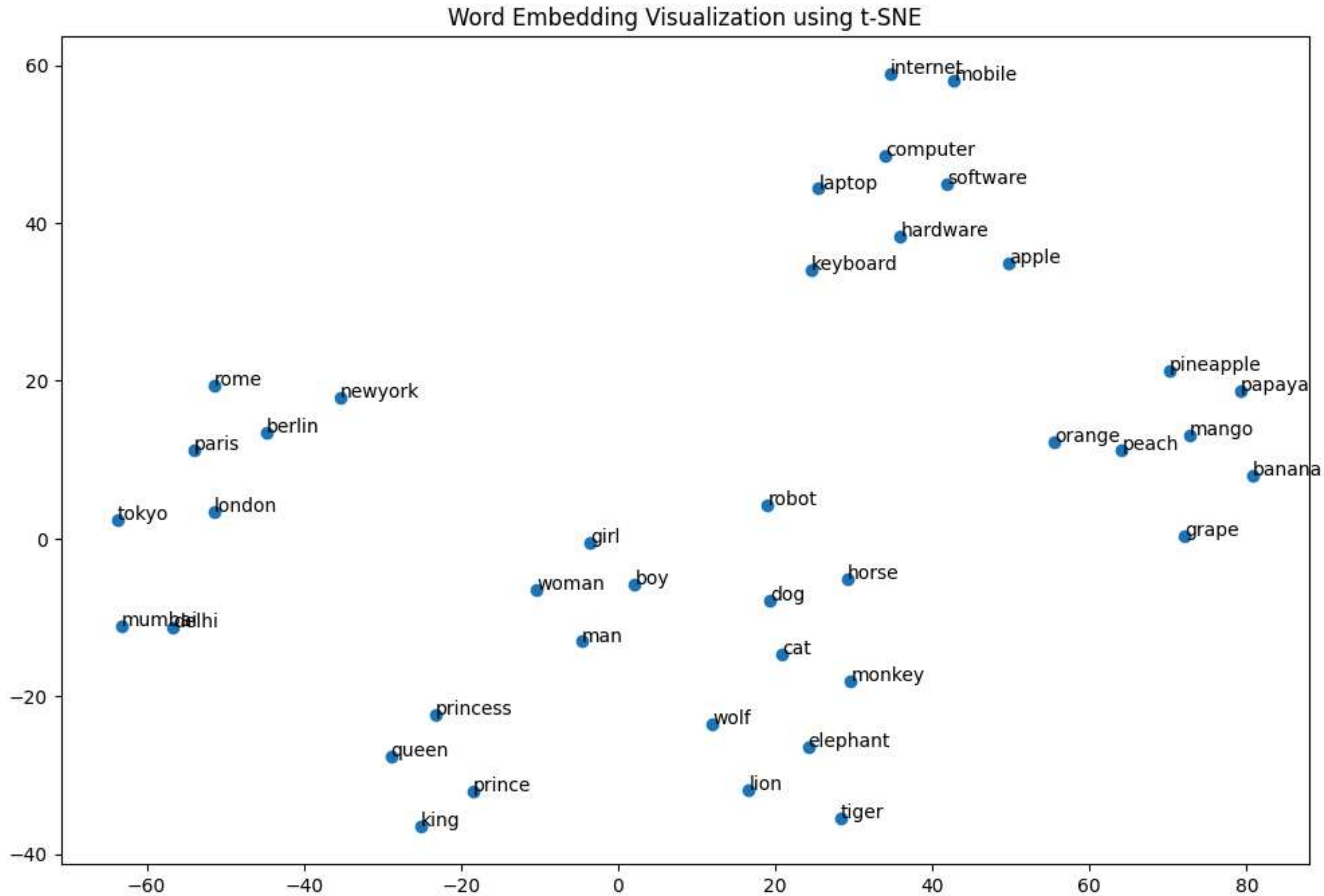
```
Shape of word vectors: (40, 100)
```

```
tsne = TSNE(n_components=2, random_state=42, perplexity=10)

reduced_vectors = tsne.fit_transform(word_vectors)

print("Shape after t-SNE:", reduced_vectors.shape)
```

```
Shape after t-SNE: (40, 2)
```

```
plt.figure(figsize=(12, 8))

x = reduced_vectors[:, 0]
y = reduced_vectors[:, 1]

plt.scatter(x, y)

# Annotate each word
for i, word in enumerate(words):
    plt.annotate(word, (x[i], y[i]))

plt.title("Word Embedding Visualization using t-SNE")
```

```
plt.show()
```


Word Embedding Visualization using t-SNE
60
40
20
0
−20
−40
−60
−40
−20
0
20
40
60
80
internet
mobile
computer
laptop
software
hardware
keyboard
apple
pineapple
papaya
orange
peach
mango
banana
grape
rome
newyork
berlin
paris
tokyo
london
robot
girl
woman
boy
horse
dog
mumbai
delhi
man
cat
monkey
princess
wolf
elephant
queen
prince
lion
tiger
king